Project Documentation

Kavya Kanaja
(ಕಾವ್ಯ ಕಣಜ)

A Comprehensive Digital Archive for Kannada Poetry

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1. Introduction

Kavya Kanaja (ಕಾವ್ಯ ಕಣಜ) is an Android application dedicated to the preservation and promotion of Kannada literature. The project serves as a modern digital repository where users can explore, listen to, and engage with Kannada poems (Bhavageethegalu) and Vachanas. It bridges the gap between traditional literary works and the digital age, making Kannada culture accessible to a global audience.

1.1 Project Objectives

- To provide a centralized platform for Kannada poetry with audio recitations.
- To offer detailed biographies of famous Kannada poets for educational purposes.
- To implement a robust offline-first architecture using Room database.
- To ensure a seamless audio experience with background playback capabilities.
- To engage users through interactive quizzes on Kannada literature.

2. Technical Specifications

The project is built using the latest Android development standards to ensure performance and scalability.

Component	Technology Used
Language	Kotlin 2.1+
UI Framework	Jetpack Compose (Material 3)
Database	Room Persistence Library

Backend	Firebase Authentication
Audio Engine	AndroidX Media3 (ExoPlayer)
Architecture	MVVM (Model-View-ViewModel)
Asynchrony	Kotlin Coroutines & Flow
Scripts	Python (ETL and Data Normalization)

3. System Architecture

The application follows the Clean Architecture pattern, emphasizing separation of concerns.

3.1 MVVM Pattern

The Model-View-ViewModel (MVVM) pattern is utilized to decouple the UI logic from the business data. The ViewModel communicates with the Repository, which handles data operations from both the Room database and external assets.

3.2 Audio Service (Foreground Service)

To provide an uninterrupted audio experience, a Foreground Service is implemented using Media3. This allows the app to continue playback even when the user navigates away or the device is locked, complying with Android's battery optimization policies.

4. Project Structure

The codebase is organized into logical packages to maintain clarity and ease of navigation:

Package	Description
com.zygisk.kavya_kanaja.auth	Firebase Authentication management logic.
com.zygisk.kavya_kanaja.data	Room entities, DAOs, and the Repository.
com.zygisk.kavya_kanaja.service	Foreground Audio Service implementation.
com.zygisk.kavya_kanaja.ui	Compose Screens, Components, and ViewModels.
app/src/main/assets	JSON metadata and core audio MP3 files.

5. Visual Interface & Features

This section showcases the user interface and functionality of the Kavya Kanaja (ಕಾವ್ಯ ಕಣಜ) project through actual application screenshots.

Figure 1: Splash Screen — App logo on launch

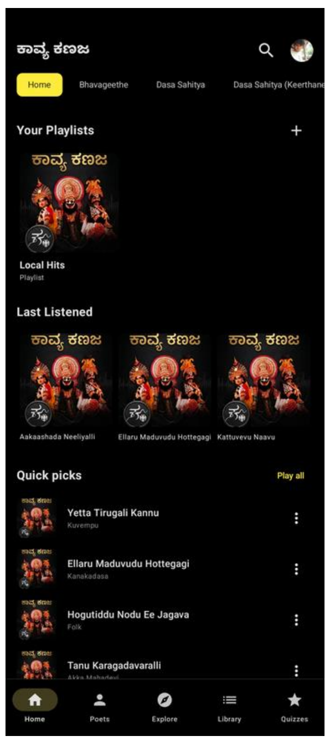

Figure 2: Home Screen — Playlists, Last Listened, and Quick Picks

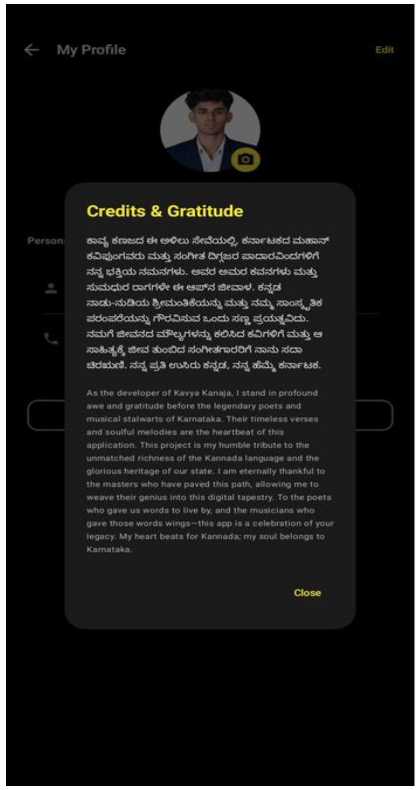

Figure 3: My Profile — Credits & Gratitude section

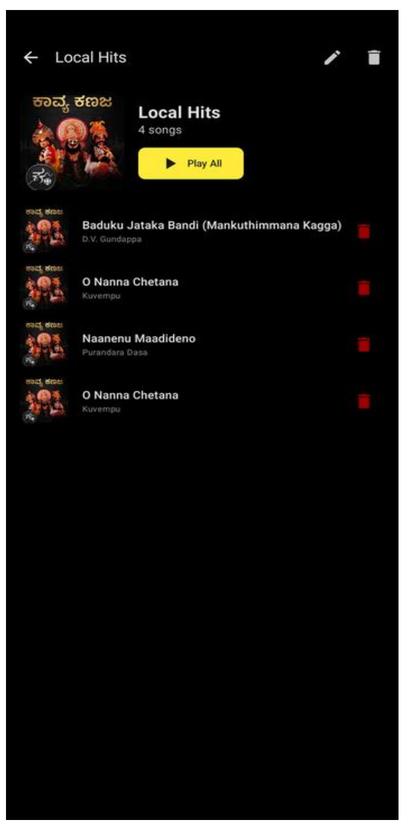

Figure 4: Playlist View — Local Hits with songs listed

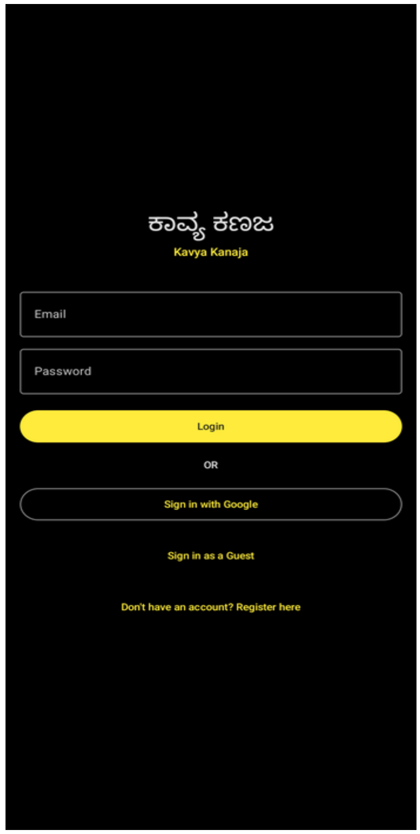

Figure 5: Login Screen — Email, Google Sign-In, and Guest access

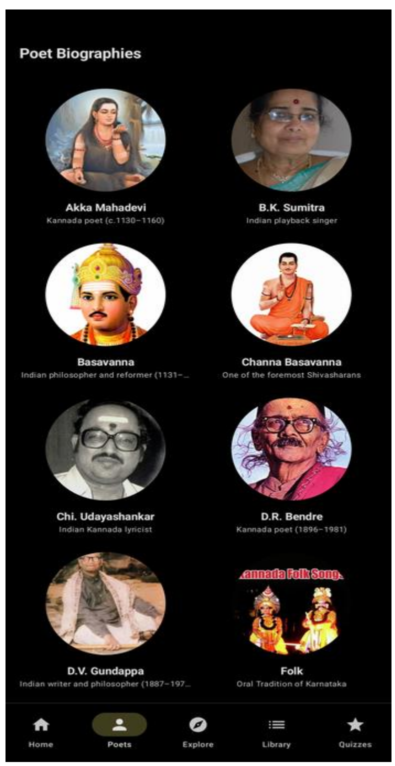

Figure 6: Poet Biographies — Grid view of featured poets

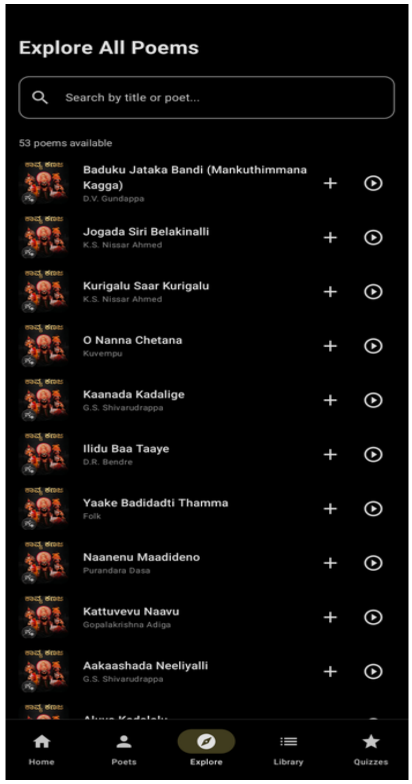

Figure 7: Explore — Browse and search all 53 poems

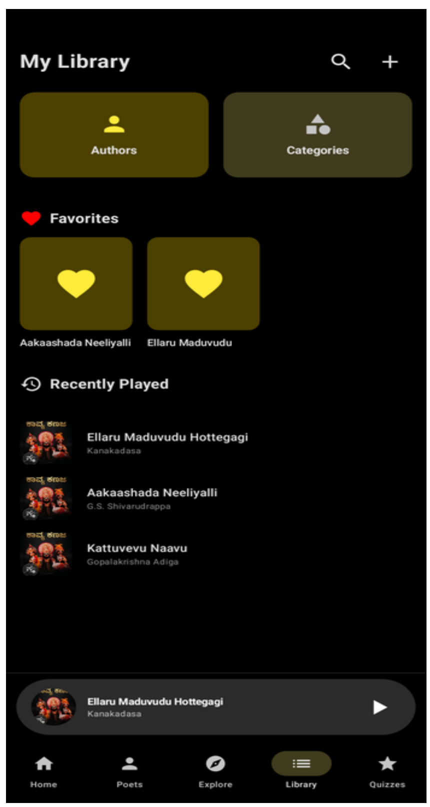

Figure 8: My Library — Favorites, Recently Played, and Categories

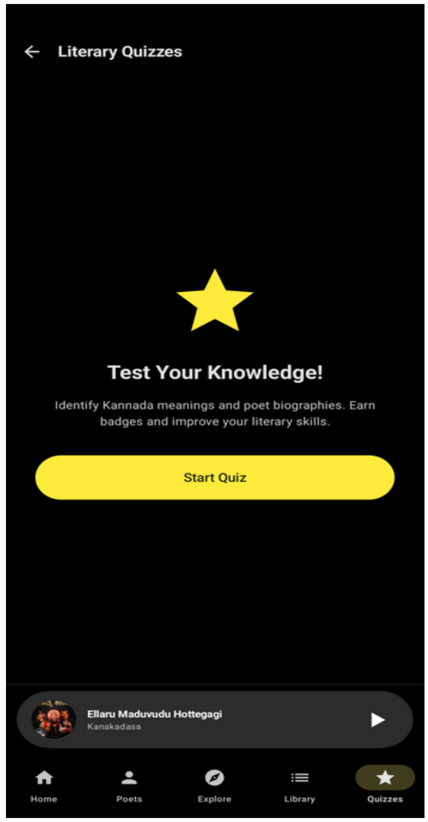

Figure 9: Literary Quizzes — Start quiz screen

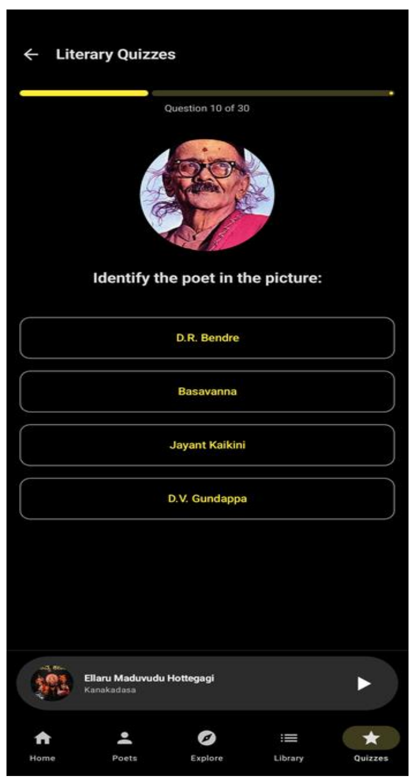

Figure 10: Quiz — Identify the poet (Question 10 of 30)

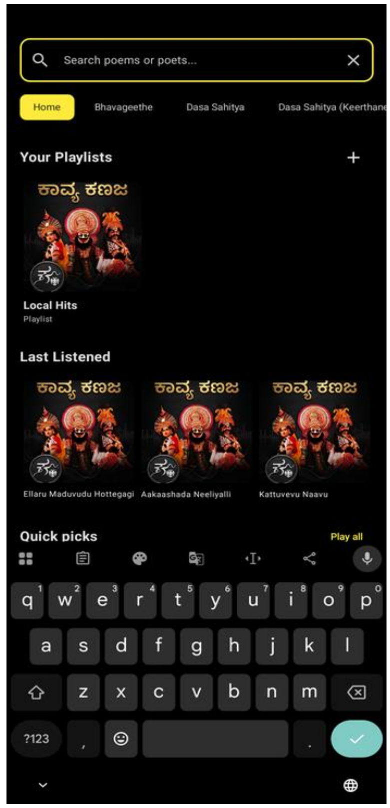

Figure 11: Search — Full-text search for poems or poets

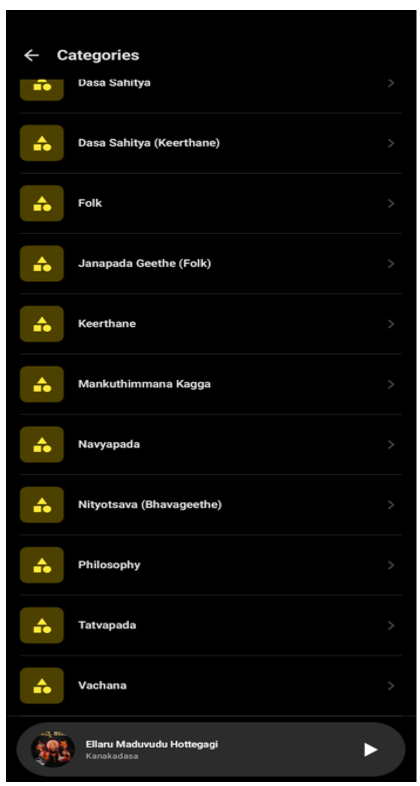

Figure 12: Categories — Dasa Sahitya, Folk, Vachana, and more

Figure 13: Poem Detail — Kannada text, word meanings, and audio player

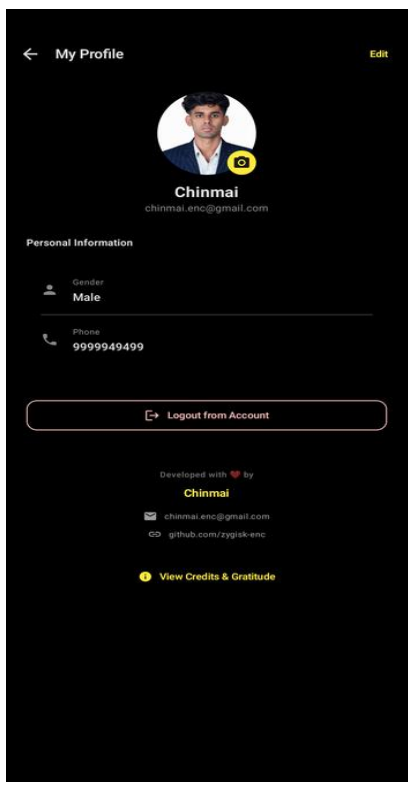

Figure 14: My Profile — User info and developer credits

Figure 15: Poet Biography — Kuvempu (Rashtrakavi 1904–1994)

6. Database Design

The application uses Room for local data persistence. Below are the primary entities defined in the schema:

Poem: Stores metadata like title, author name, category, and audio source path.
PoetBio: Contains detailed biographies, birth details, and award information for authors.
Playlist: User-created collections of poems.
PoemProgress: Tracks the listening progress and last played timestamp for each poem.
UserProfile: Stores user-specific metadata like display name and profile image reference.

7. Data Management & Automation

To maintain a high-quality content registry, the project includes several Python utility scripts for back-end data management:

bulk_update_poems.py: Automates the synchronization of metadata between the assets and the database.

normalize_poems.py: Ensures consistent naming conventions and character encoding for Kannada text.

download_images.py: Fetches poet avatars and assets from verified sources.

8. Conclusion

Kavya Kanaja (ಕಾವ್ಯ ಕಣಜ) successfully demonstrates the integration of modern Android development techniques with cultural preservation goals. By utilizing Jetpack Compose for the UI, Room for persistence, and Media3 for audio, the project provides a robust foundation for a community-driven literary platform. Future iterations could include community poem uploads, social sharing features, and deeper AI-driven content recommendations.

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